AIH GP: 2A Group Proposal (6 February 2026)

Members:

1. Rachel Lau
2. Kainoa Martinez
3. Jingran Zheng
4. Sarah Lee
5. XinnuanWang

Communication Plan:

- Weekly Meetings: In-Person, every Thursday after AIH GP class (~ 1 hour)
- WhatsApp Group Chat: Constant communication
- GitHub Project Board: Communicate task division and deadlines

Division of Roles:

Role	Name(s)	Responsibilities
Project Manager	Kainoa	● Ensure deadlines are met. ● Ensure all work is completed on time and according to our internal deadlines. ● Ensure all meeting objectives are met. ● Maintain the Github project board
Technical Lead	Rachel	● Lead the development and implementation of the NLP model ● Ensure model meets MVP and optimal product targets ● Model training and optimization
Data Lead	Jingran	● Lead data preprocessing and cleaning ● Manages TRE data access and handling
Evaluation & Testing Lead	Sarah	● Lead the testing of model performance and validation on TRE real data ● Quality assurance
Code Team	~~Everyone~~	● Involved with the development of the model (including the coding and technical aspects) ● Design and determine evaluation framework ● Select appropriate NLP architecture

		(e.g., BERT vs alternatives)
Research Lead (s)	Xinnuan	• Consolidate all the background research and literature review conducted by the team • Summarize all the available literature and evidence available
Research Team	Everyone	• Conduct background research and literature review • Literature review on NLP for clinical text • Research on difficulty prediction approaches

**All aspects of the project will be collaboratively completed by the team. The responsibility of the managers and leads is to lead the development of their respective sections and ensure that their section is developing according to the timeline.*

Project Outline 1.0

Project Brief: Using clinical reports free text to determine the level of difficulty for computer vision models.

Project Aim: Our group aims to use synthetic data to develop a machine learning model, before training and testing the model on the real data to predict the level of difficulty for computer vision models.

Project Objectives:

- Conduct sufficient literature review and background research to understand the current evidence on the use of natural language processing (NLP) on clinical reports free text.
- Obtain and choose an appropriate synthetic dataset to develop the model on.
- Explicitly define what constitutes "difficulty" and how it will be measured/labeled.
- Determine the appropriate NLP model (e.g. Transformer based models) that best meets the optimal product we aim to achieve.
- Develop a NLP model that has robust performance, incorporates multiple approaches and is able to provide confidence scores for predictions.
- Obtain a better understanding of how to use TRE by training and testing the developed model on real data using the TRE

Minimum Viable Product:

- Develop a working NLP model that predicts level of difficulty from clinical report free text with ≥70% accuracy.
- Incorporate a single approach to the model.
- Goal: Proof of concept for our model and that our model works.

Optimal Product:

- High-accuracy model (≥85%) with robust performance when trained and tested on real clinical data.
- Incorporates multiple approaches (ensemble methods, transformer models like BERT).
- Able to handles edge cases and missing data gracefully;
- If possible, the model should be able to provide confidence scores for predictions.

Brief Task Breakdown:

Task	Deadline
Background and literature review	Week 5
Find the appropriate synthetic dataset	Week 5
Decide what NLP model to use	Week 6
Develop the model	Week 7
Train and test developed model with real data in TRE	Week 8/9

Individual Group Write Ups	• 11 Feb • 11 Mar • 1 Apr
Group Presentation	26 March 2026
Completed Group Report	8 May 2026

Project Outline 2.0

Project Title: Using clinical reports free text to determine the level of difficulty for computer vision models.

Project Aim: Our group aims to use synthetic data to develop a machine learning model, before training and testing the model on the real data to predict the level of difficulty for computer vision models.

Before implementation, we need to decide how we define 'difficulty'.

- Level of Difficulty: Predict how difficult an image will be for a CV model using report text
    - Ground Truth (Targets): Difficulty labels derived from CV model performance on images (Easy/Moderate/Hard based on confidence scores and accuracy)
    - Predictions: Difficulty categories predicted by our NLP model using features extracted from radiology reports
    - Evaluation: Compare NLP model predictions to ground truth difficulty labels to measure accuracy
- Image Type: Chest X-rays
- Chosen CV Model:
    - TorchXRayVision or CheXNet
- Chosen NLP:
    - CheXpert labeler
    - Clinical-BERT
- Chosen ML Model:
    - Random Forest
    - XGBoost
    - Neural Network
- Chosen Training Dataset: IU Chest X-rays

Minimum Viable Product:

- Develop a working NLP model that predicts level of difficulty from clinical report free text with ≥70% accuracy.
    - Accuracy: Compare NLP model predictions to ground truth difficulty labels (obtained via comparing confidence score and predictions to some predefined threshold)
- Incorporate a single approach to the model.
    - Start with Random Forest and XGBoost and compare accuracy for the better model
- Goal: Proof of concept for our model and that our model works.

Optimal Product:

- High-accuracy model (≥85%) with robust performance when trained and tested on real clinical data.
- Incorporates multiple approaches (ensemble methods, transformer models like BERT).
    - Hybrid Feature Ensemble: Combining different types of features from various sources (not just transformers)
- Able to handles edge cases and missing data gracefully
- If possible, the model should be able to provide confidence scores for predictions.

POSSIBLE PROJECT FLOW

PHASE 1: Establish Ground Truth Difficulty

---

1. Run CV model on medical images
   └→ Get: predictions + confidence scores for each case

2. Label each case by difficulty using thresholds

Difficulty Thresholds:

- EASY: confidence > 0.85 AND correct prediction
- HARD: confidence < 0.60 OR incorrect prediction
- MODERATE: everything else

PHASE 2: Extract NLP Features from Reports

---

3. Extract features from radiology reports:
    - Option A: CheXpert labeler (uncertainty counts)
    - Option B: Clinical-BERT (embeddings)
    - Option C: Both (hybrid - RECOMMENDED)

PHASE 3: Train Text-Based Difficulty Predictor

---

4. Train classification model:
    - Input: NLP features from reports
    - Output: Difficulty category (Easy/Moderate/Hard)
    - Model: Train both a RF and XGBoost model and compare which model performs better

---

PHASE 4: Evaluate and Validate

---

5. Compare predictions to ground truth difficulty labels
    - Metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix
6. Analyze: Do report features actually predict CV model difficulty?

NEXT STEPS

~~Threshold Selection (Step 2):~~

- Try multiple thresholds and see which gives best class balance:
- Option 1: Confidence-based
    - EASY: confidence > 0.85 and correct
    - HARD: confidence < 0.60 or incorrect
- Option 2: Percentile-based (ensures balanced classes)
    - EASY: Top 33% confidence + correct
    - HARD: Bottom 33% confidence or incorrect
    - MODERATE: Middle 33%

~~Feature Selection (Step 3):~~

- Start with CheXpert (interpretable, fast)
- Add Clinical-BERT if accuracy is insufficient
- Compare both approaches in your results

~~Model Selection (Step 4):~~

- Random Forest: Good baseline, shows feature importance
- XGBoost: Often better performance
- Neural Network: Best for Clinical-BERT embeddings

Improvements?

- Binary classification instead, Easy or Hard.
- The task we are setting and the method we are doing is a bit disjointed. Doesn't directly answer our question.
- Redefine what the CV model and NLP model is trying to do and what is the linkage?
- CV model – Classifying chest X-ray findings and outputting a confidence score for each case
- NLP model – Reading the paired radiology report and predicting whether that confidence score will be high or low. NLP model looks for keywords and linguistic patterns in the report associated with high or low CV confidence. Input = NLP features extracted from reports, Output = Predicted difficulty label (High confidence/Low confidence)

EASY: >0.75 – The image is straightforward for the CV model
HARD: < 0.60 – The image is difficult for the CV model

Project Outline 3.0 (Final)

Project Aim:
Develop a machine learning model to predict chest X-ray image difficulty for computer vision (CV) classifiers using radiology report text.

- Difficulty is defined as how difficult it is for a CV model to correctly classify an image
- This system would enable prioritisation of difficult cases for expert human review, improving clinical workflow efficiency and resource allocation.

Research question:
Do radiology free-text reports contain sufficient linguistic information to predict image classification difficulty for CV models?

Overview:
The model pipeline consists of 3 components:

1. CV model
    1. Generates difficulty labels (ground truth targets)
2. NLP
    1. Extracts features from radiology reports (input features)
3. ML classifier
    1. Predicts difficulty from text features (extracted using NLP)

Computer Vision Model: (Ground truth generation)

- Purpose: Generate difficulty labels for training data
- Model: TorchXRayVision DenseNet-121
    - Pre-defined pathologies used
    - Outputs confidence scores (0-1) for 14 pathology classes
    - Confidence scores represent the model's certainty regarding the presence of specific thoracic pathologies in a chest X-ray image
- Confidence score extraction:
    - Each image produces 14 pathology confidence scores
    - Max confidence scores are obtained and used as the overall model certainty metric
- Difficulty label assignment:
    - Fixed threshold: Based on the confidence score obtained, the image is classified as either ‘easy’ or ‘hard’ to classify
        - EASY: confidence score >= 0.75 - suggests that the image is straightforward for the CV model to classify

        - HARD: confidence score < 0.75 - suggests that the image is difficult for the CV model to classify
    - If classes are imbalanced, we switch to a percentile-based assignment to ensure balanced classes

Natural Language Processing: (Feature extraction)

- Purpose: Extract semantic features from radiology reports
- Input Data:
    - Radiology free-text reports corresponding to X-ray images
    - Matches to images
- Model: Clinical-BERT (Bio_ClinicalBERT)
    - Transformer-based language model pre-trained on clinical text
    - Produces 768-dimensional semantic embeddings
- Pipeline:
    - (1) Combine findings + impression sections from each report
    - (2) Tokenise text
    - (3) Pass through Clinical-BERT encoder
    - (4) Extract [CLS] token embedding
        - This embedding captures semantic meaning of clinical text
- Hypothesis: Report text contains linguistic signals of difficult cases

Machine Learning Classifier: (Difficulty prediction)

- Purpose: Predict image difficulty from report embeddings
- Input Features (X): 768-dimensional BERT embeddings
- Target Labels (Y): Difficulty categories (EASY vs HARD)
- Model: XGBoost Gradient Boost Classifier
- Evaluation Metrics:
    - Confusion matrix
    - F1-score
    - Accuracy
    - Precision & Recall
- Success Criterion: F1 > 0.7 (substantial improvement over baseline)
- Binary Classification Targets
    - NEGATIVE_RESULT = "F1 < 0.55" # No meaningful signal
    - REALISTIC_TARGET = "F1 = 0.60-0.70" # Demonstrates feasibility
    - OPTIMISTIC_TARGET = "F1 = 0.70-0.80" # Strong performance
    - SUCCESS_THRESHOLD = "F1 ≥ 0.65" # Minimum for practical value